Barriers to Persistence in Competitive Programming Training Among Female Students: Why Many Start, But Few Continue

acm International Collegiate Programming Contest

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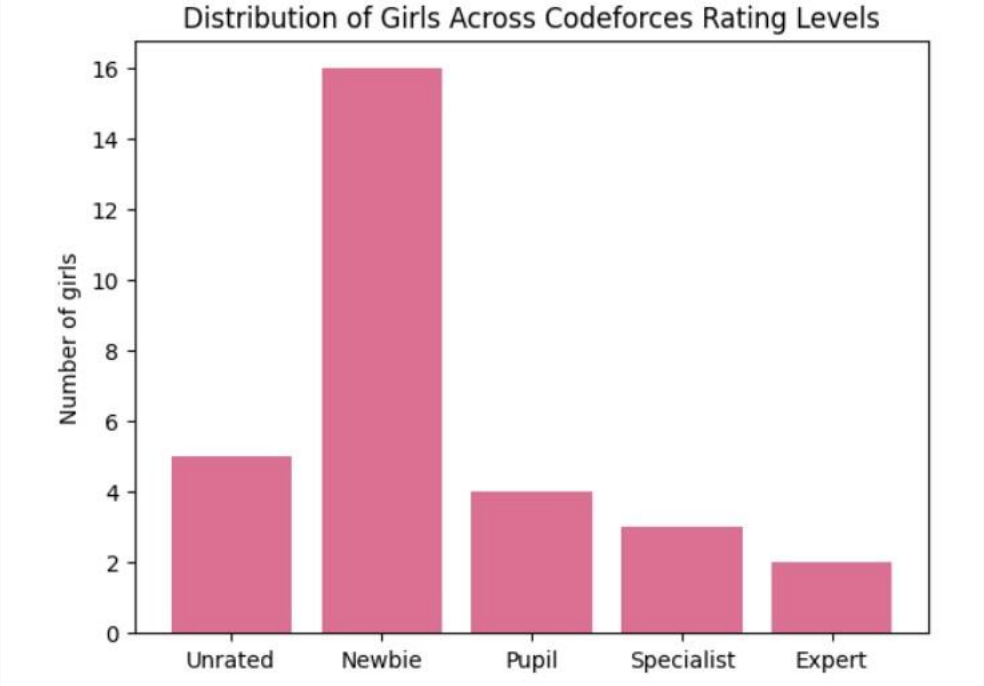
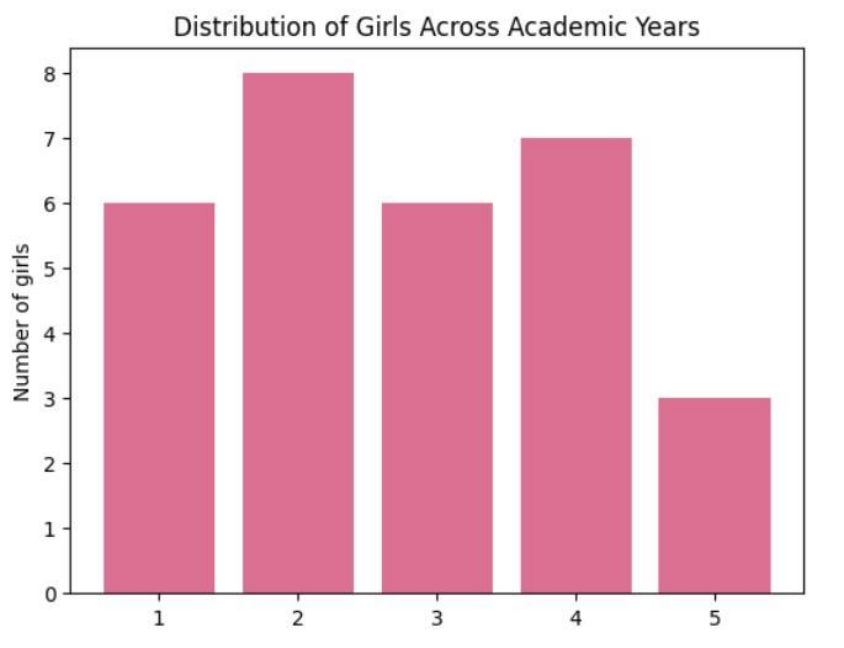
Distribution of Girls Across Rating Levels



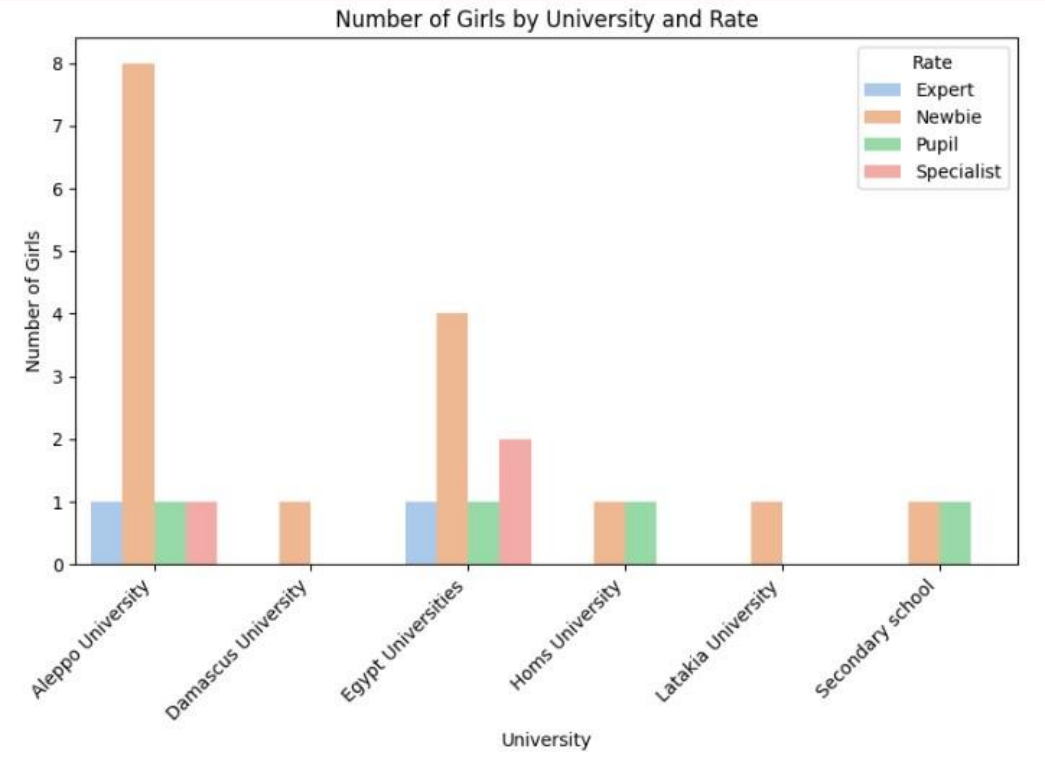
The analysis shows that most girls fall into the **Unrated and Newbie categories**, while fewer students reach **Pupil, Specialist, or Expert levels**.

Distribution Across Academic Year

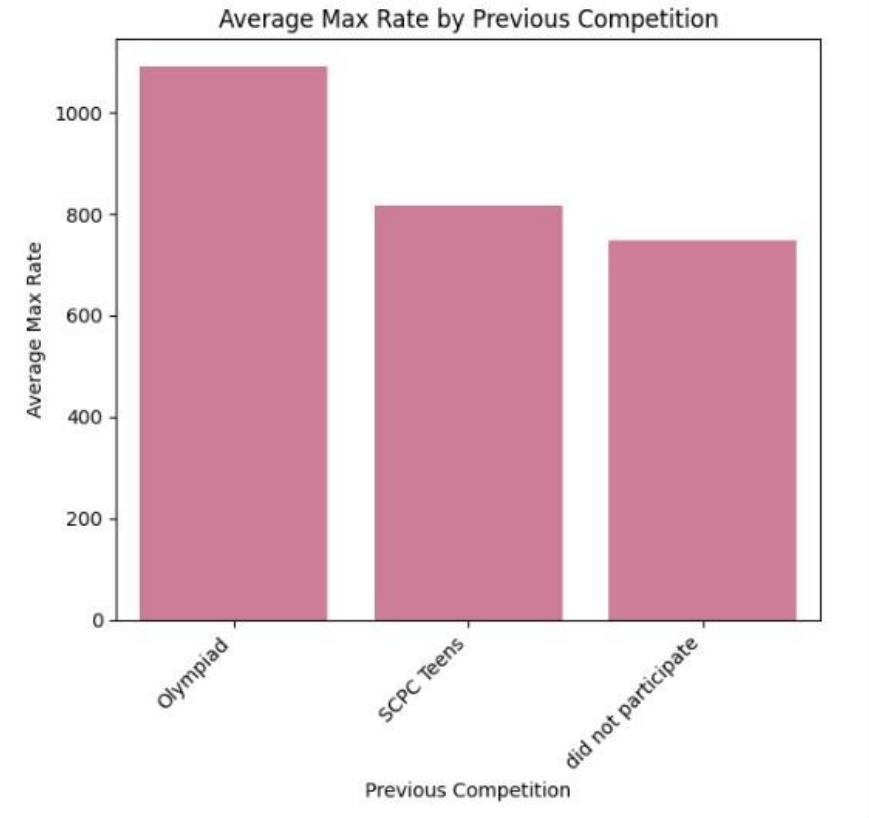


The data shows that most participants are from **early academic years**, while fewer girls are in advanced years.

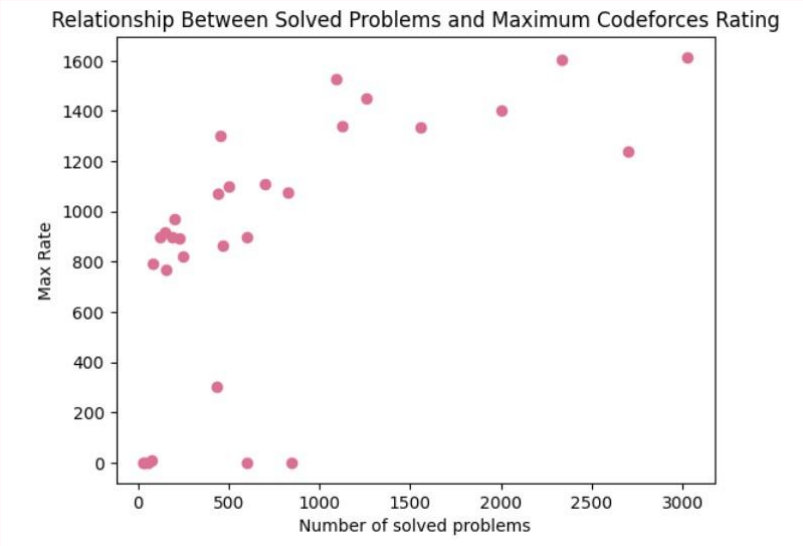
University & Rating Comparison:



Average Max Rate by Previous Competition



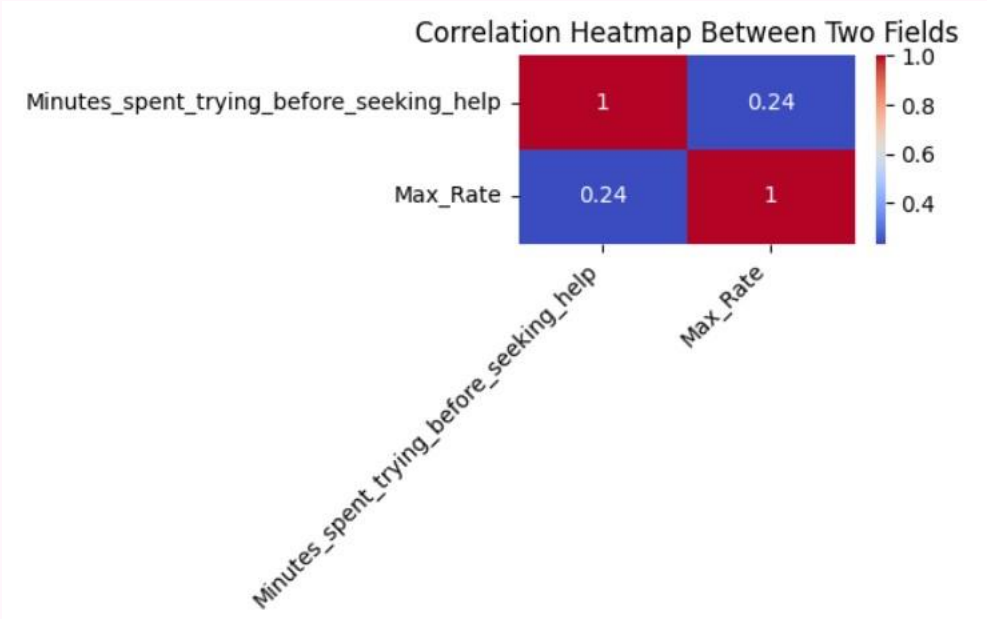
Relationship Between Solved Problems and Max Rate



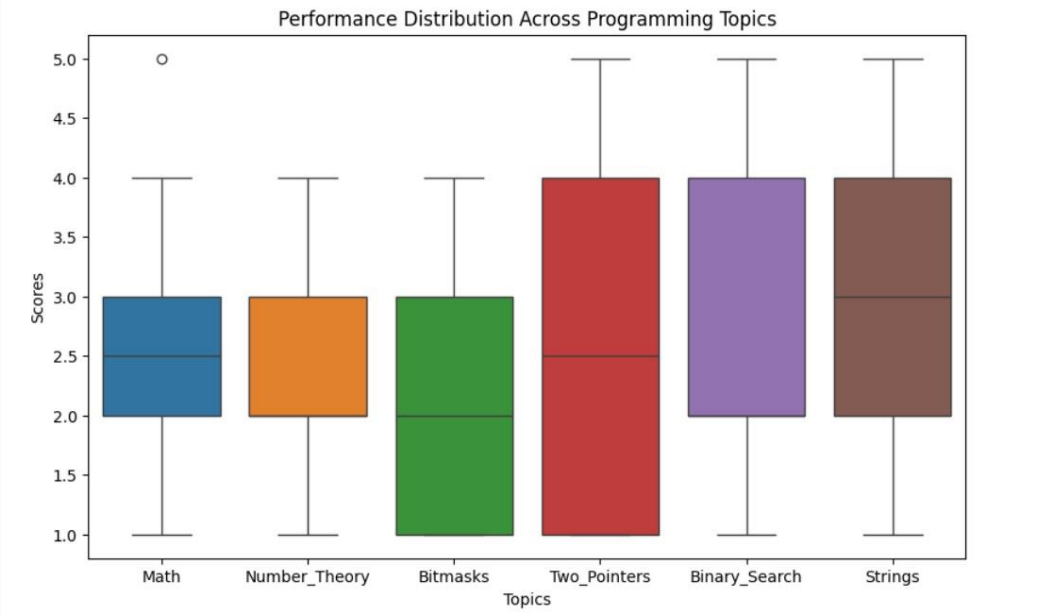
The scatter plot shows a **positive relationship between the number of solved problems and maximum rating**.
Students who solve more problems tend to achieve higher Codeforces ratings.

Correlation Analysis

A moderate positive relationship was observed between the time spent trying before asking for help and the highest rating on Codeforces. For reasons, this relationship may be affected by several confounding factors, including lack of participation in Codeforces contests, irregular training schedules, and others.

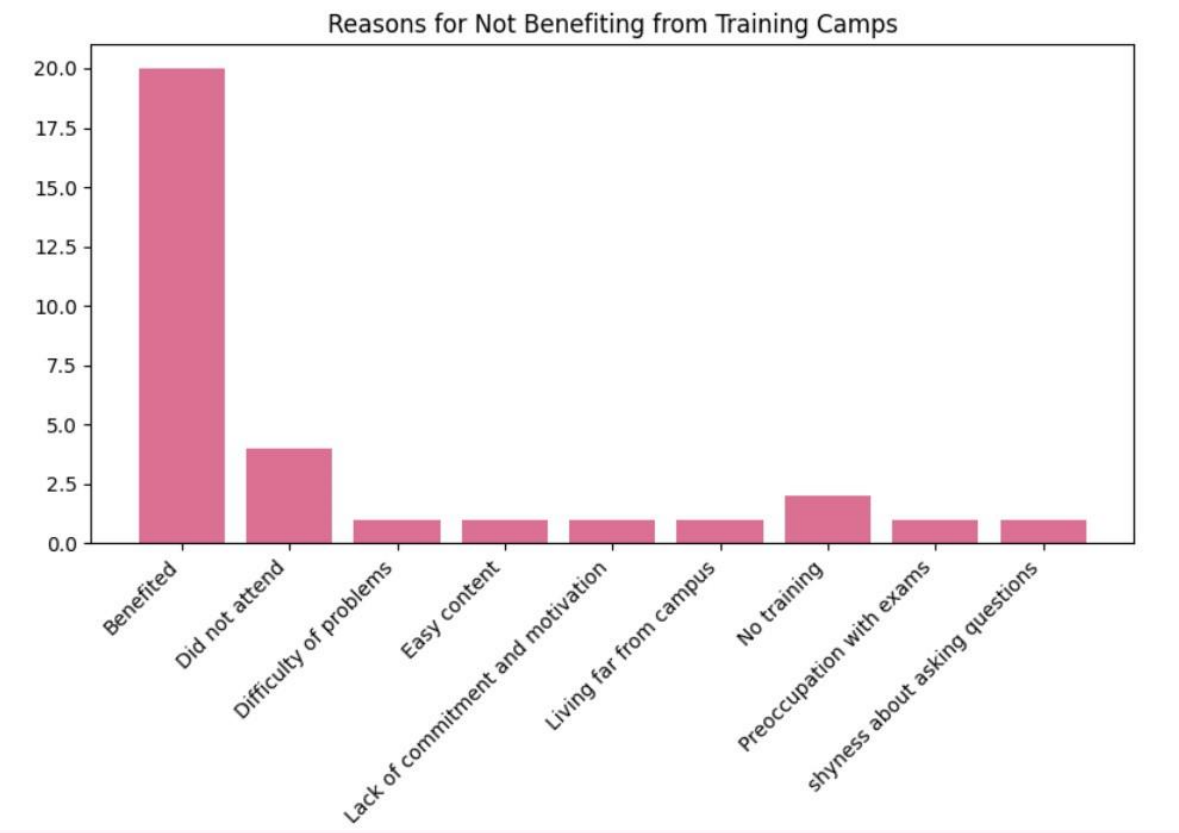


Topic Strength Analysis

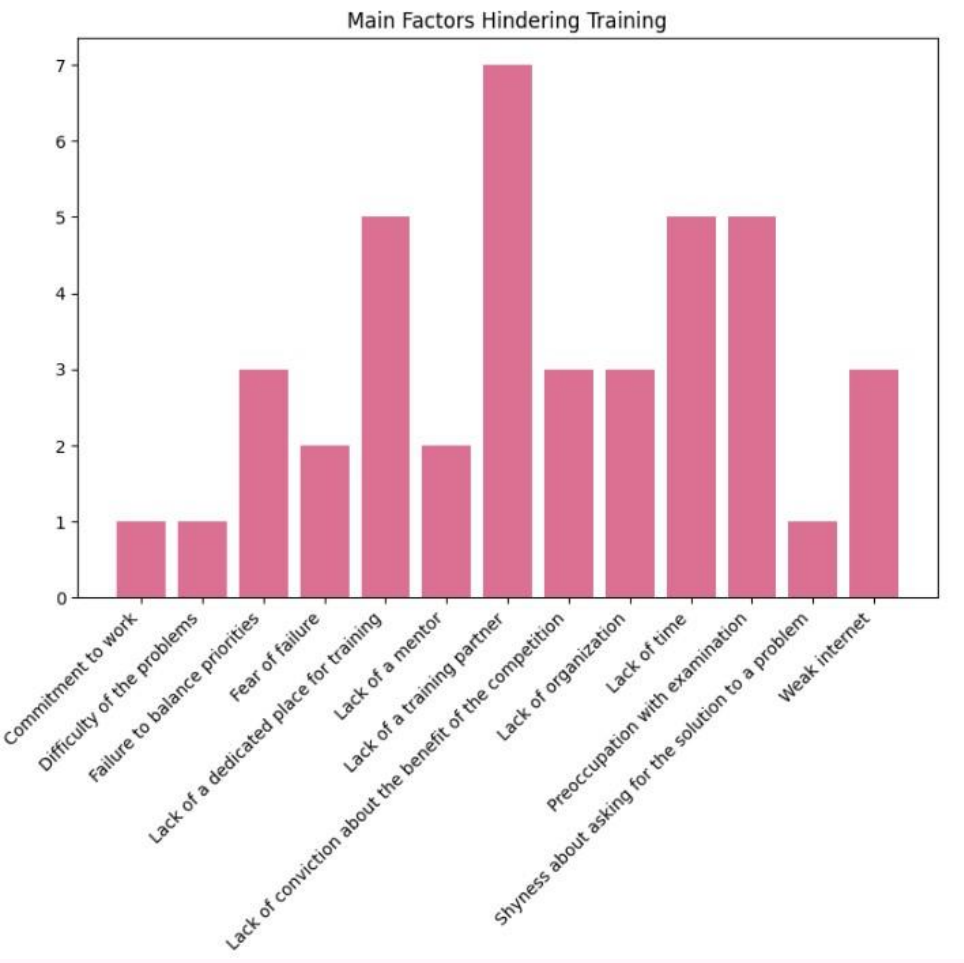
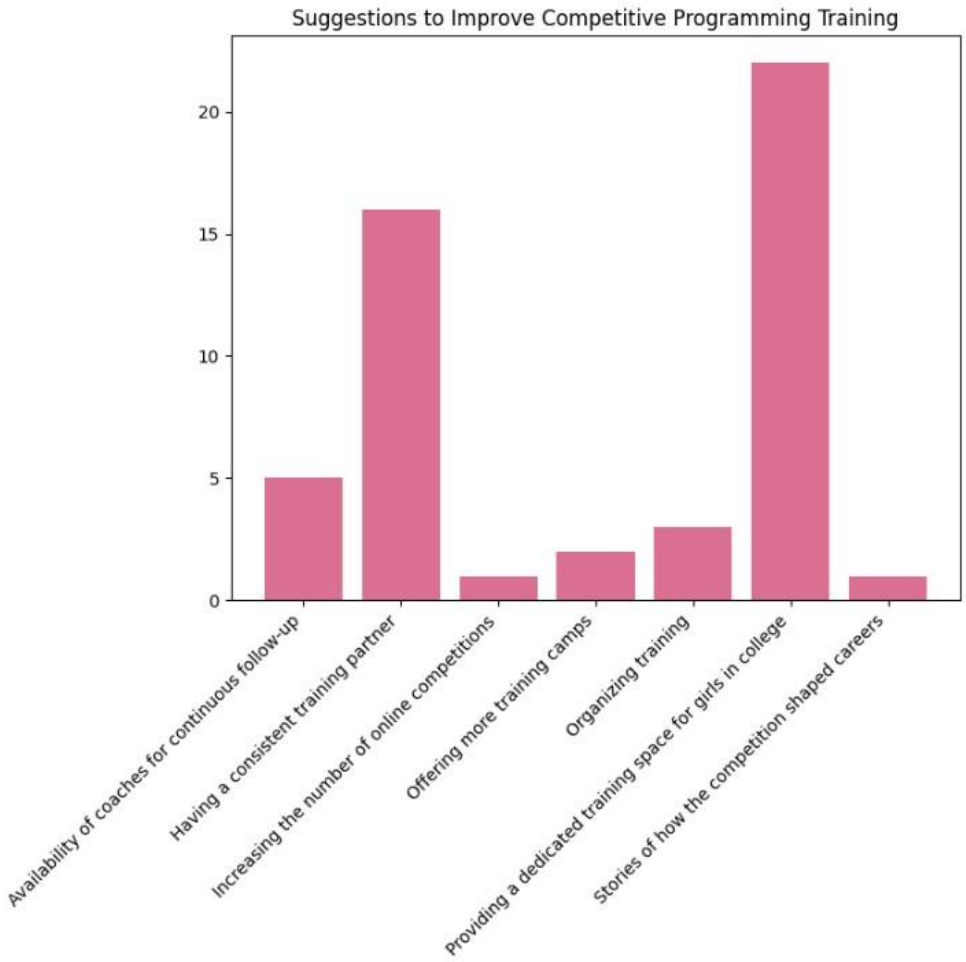


Students performed best in **Strings** and **Binary Search** topics.
Bitmasks showed slightly lower average scores, suggesting this area may require more focus.

Reasons for Not Benefiting from Training Camps



Reasons Hindering Training



Suggestions to Improve Training